

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.tree import DecisionTreeRegressor

from sklearn.metrics import mean_squared_error, r2_score
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_s

# Load dataset

df = pd.read_csv("/content/superstore.csv")

```

```
# Display first few rows
```

```
df.head()
```

	Category	City	Country	Customer.ID	Customer.Name	Discount	Market	记录数
0	Office Supplies	Los Angeles	United States	LS-172304	Lycoris Saunders	0.0	US	1
1	Office Supplies	Los Angeles	United States	MV-174854	Mark Van Huff	0.0	US	1
2	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	0.0	US	1
3	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	0.0	US	1
4	Office Supplies	Los Angeles	United States	AP-109154	Arthur Prichep	0.0	US	1

5 rows × 27 columns

```
df.shape
```

```
(51290, 27)
```

```
# Basic info about dataset
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 27 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Category              51290 non-null  object
1   City                  51290 non-null  object
2   Country              51290 non-null  object
3   Customer.ID          51290 non-null  object
4   Customer.Name        51290 non-null  object
5   Discount              51290 non-null  float64
6   Market               51290 non-null  object
7   记录数                51290 non-null  int64
8   Order.Date           51290 non-null  object
9   Order.ID             51290 non-null  object
10  Order.Priority       51290 non-null  object
11  Product.ID           51290 non-null  object
12  Product.Name         51290 non-null  object
13  Profit               51290 non-null  float64
14  Quantity             51290 non-null  int64
15  Region               51290 non-null  object
16  Row.ID               51290 non-null  int64
17  Sales                51290 non-null  int64
18  Segment              51290 non-null  object
19  Ship.Date            51290 non-null  object
20  Ship.Mode            51290 non-null  object
21  Shipping.Cost        51290 non-null  float64
22  State                51290 non-null  object
23  Sub.Category         51290 non-null  object
24  Year                 51290 non-null  int64
25  Market2              51290 non-null  object
26  weeknum              51290 non-null  int64
dtypes: float64(3), int64(6), object(18)
memory usage: 10.6+ MB
```

```
df.columns
```

```
Index(['Category', 'City', 'Country', 'Customer.ID', 'Customer.Name',
      'Discount', 'Market', '记录数', 'Order.Date', 'Order.ID',
      'Order.Priority',
      'Product.ID', 'Product.Name', 'Profit', 'Quantity', 'Region', 'Row.ID',
      'Sales', 'Segment', 'Ship.Date', 'Ship.Mode', 'Shipping.Cost', 'State',
      'Sub.Category', 'Year', 'Market2', 'weeknum'],
      dtype='object')
```

```
# Summary statistics
df.describe()
```

	Discount	记录数	Profit	Quantity	Row.ID	Sale
count	51290.000000	51290.0	51290.000000	51290.000000	51290.00000	51290.00000
mean	0.142908	1.0	28.610982	3.476545	25645.50000	246.49844
std	0.212280	0.0	174.340972	2.278766	14806.29199	487.56717
min	0.000000	1.0	-6599.978000	1.000000	1.00000	0.00000
25%	0.000000	1.0	0.000000	2.000000	12823.25000	31.00000
50%	0.000000	1.0	9.240000	3.000000	25645.50000	85.00000
75%	0.200000	1.0	36.810000	5.000000	38467.75000	251.00000
max	0.850000	1.0	8399.976000	14.000000	51290.00000	22638.00000

```
# Check missing values
df.isnull().sum()
```

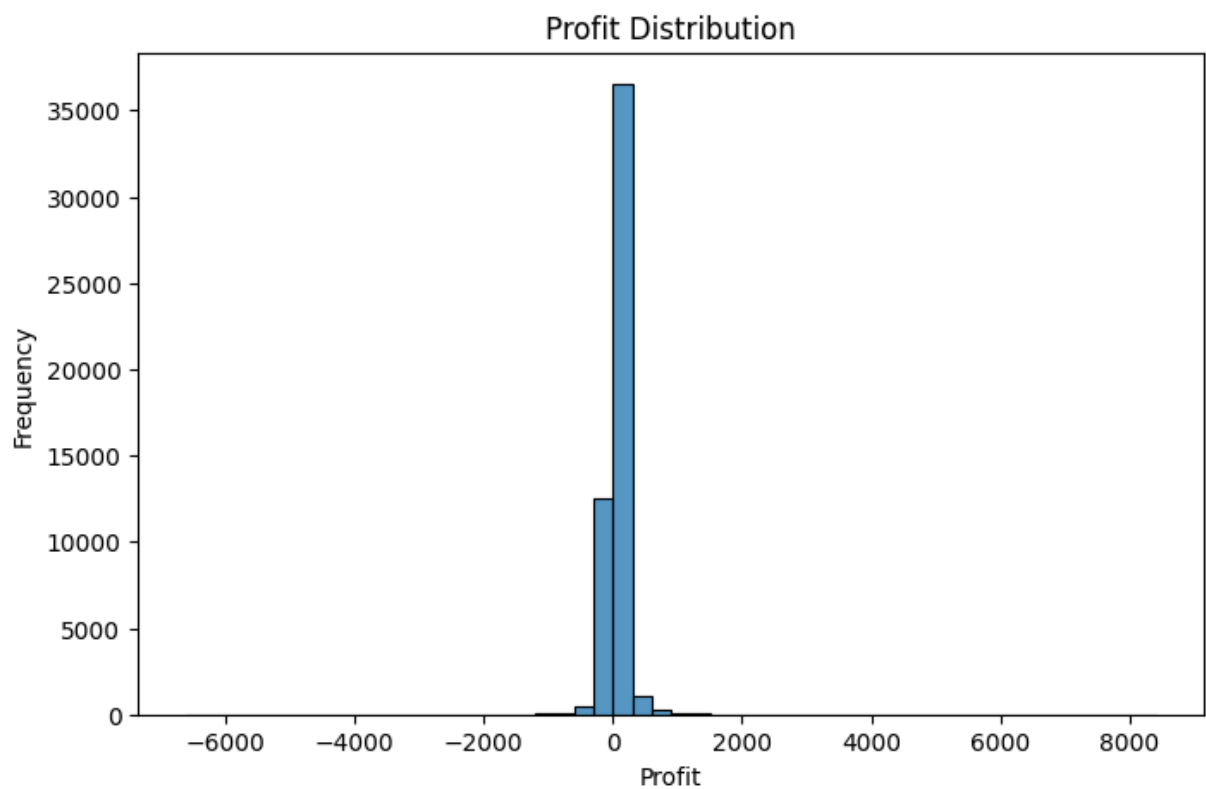
	0
Category	0
City	0
Country	0
Customer.ID	0
Customer.Name	0
Discount	0
Market	0
记录数	0
Order.Date	0
Order.ID	0
Order.Priority	0
Product.ID	0
Product.Name	0
Profit	0
Quantity	0
Region	0
Row.ID	0
Sales	0
Segment	0
Ship.Date	0
Ship.Mode	0
Shipping.Cost	0
State	0
Sub.Category	0
Year	0
Market2	0
weeknum	0

dtype: int64

```
# Remove duplicates
df = df.drop_duplicates()
```

```
df = df.drop(columns=['记录数'])
```

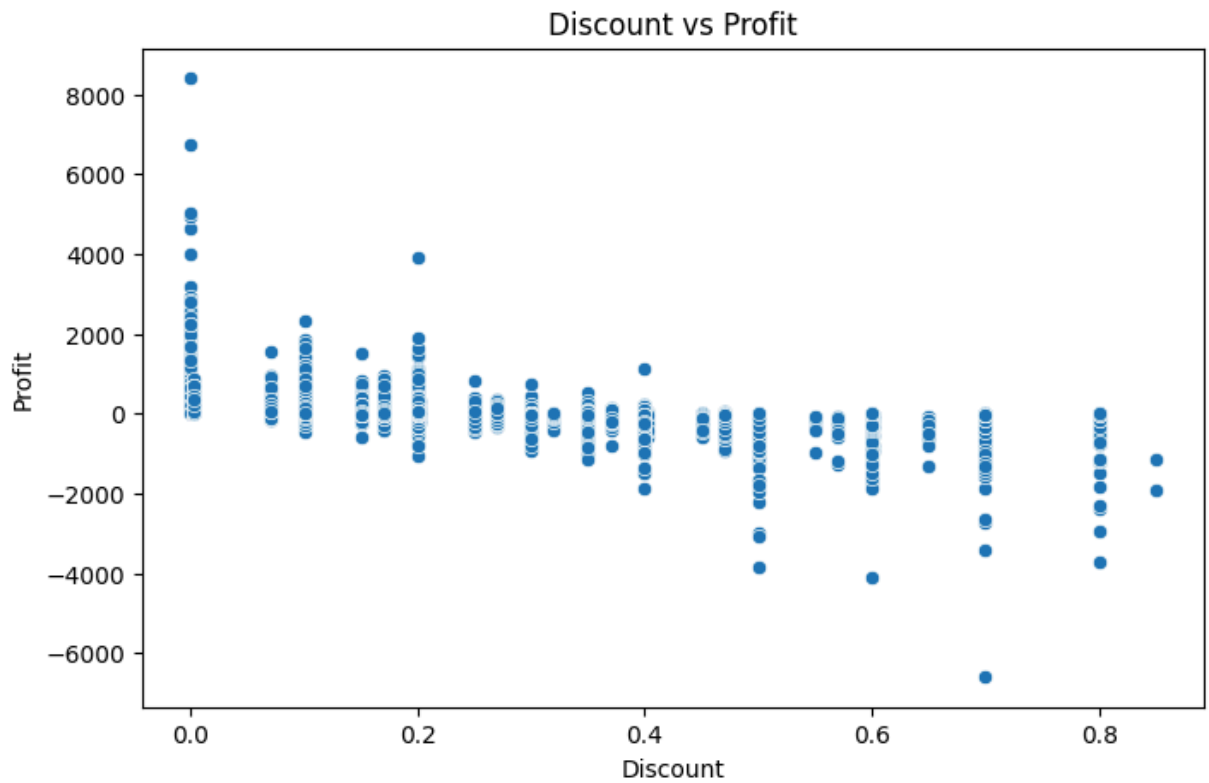
```
plt.figure(figsize=(8,5))
sns.histplot(df['Profit'], bins=50)
plt.title("Profit Distribution")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.show()
```



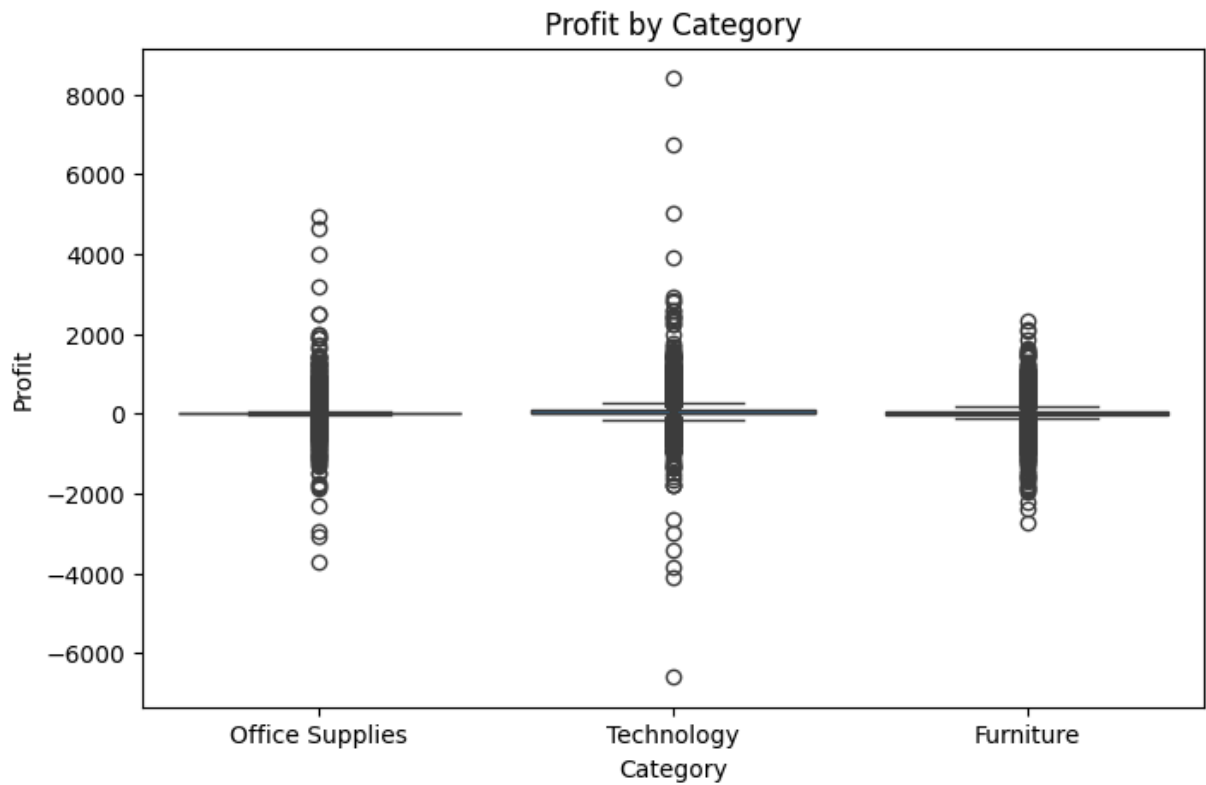
```
plt.figure(figsize=(8,5))
sns.scatterplot(x='Sales', y='Profit', data=df)
plt.title("Sales vs Profit")
plt.show()
```



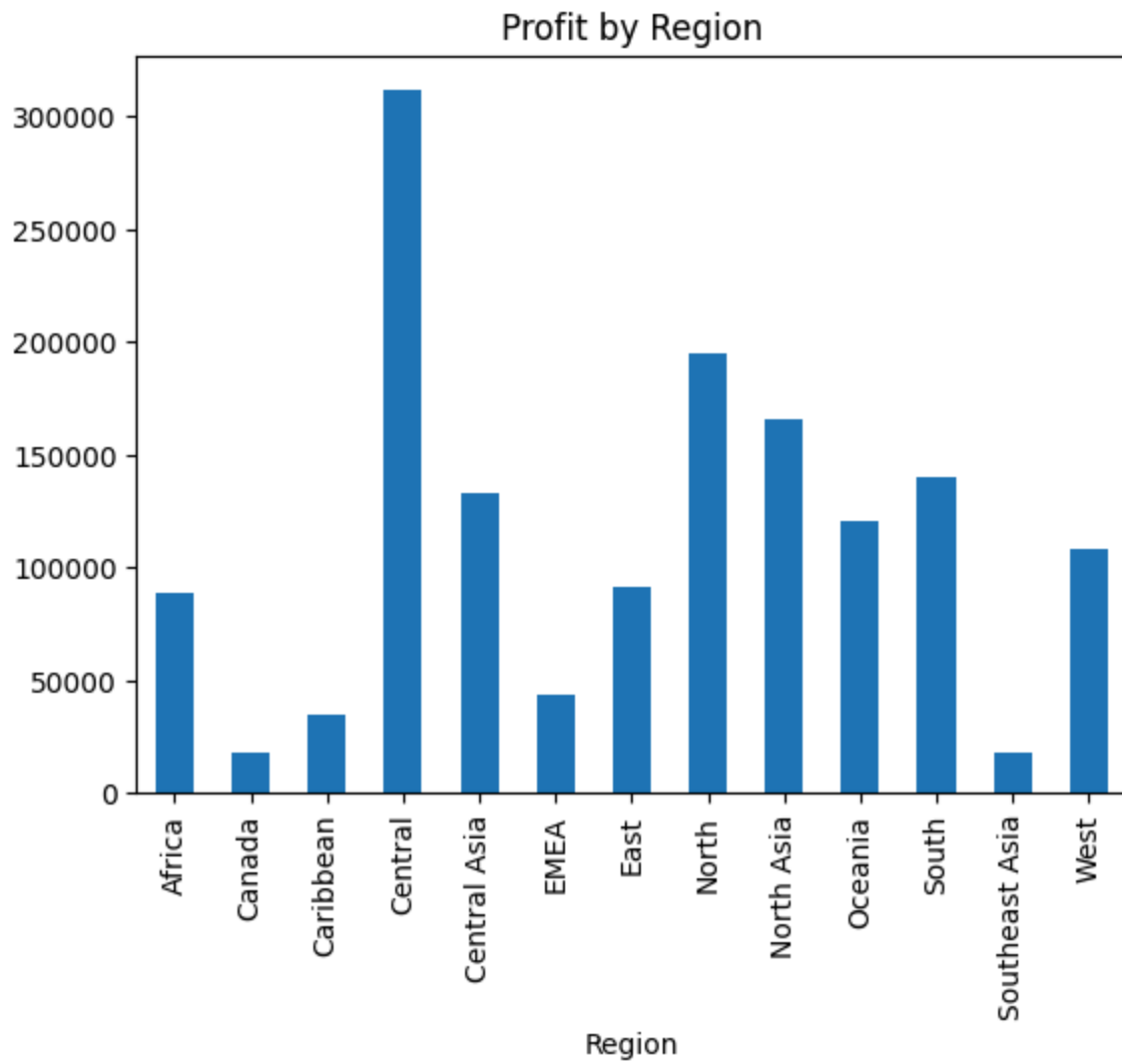
```
plt.figure(figsize=(8,5))  
sns.scatterplot(x='Discount', y='Profit', data=df)  
plt.title("Discount vs Profit")  
plt.show()
```



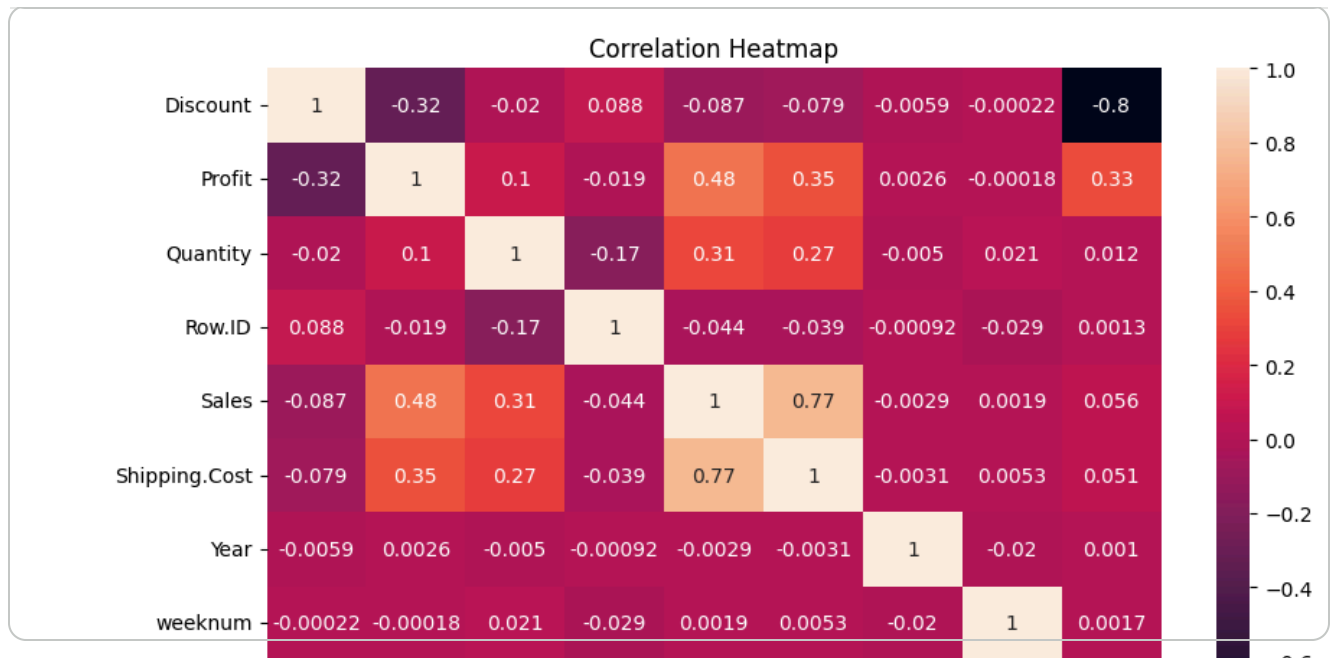
```
plt.figure(figsize=(8,5))
sns.boxplot(x='Category', y='Profit', data=df)
plt.title("Profit by Category")
plt.show()
```



```
df.groupby('Region')['Profit'].sum().plot(kind='bar')  
plt.title("Profit by Region")  
plt.show()
```

```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



```
X = df[['Sales', 'Discount', 'Quantity']]
y = df['Profit']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()

model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

```
# Calculate RMSE

mse = mean_squared_error(y_test, predictions)
rmse = np.sqrt(mse)

r2 = r2_score(y_test, predictions)

print("Linear Regression RMSE:", rmse)
print("Linear Regression R2:", r2)
```

```
Linear Regression RMSE: 156.32231070810542
Linear Regression R2: 0.1682953647569665
```

```
tree = DecisionTreeRegressor(random_state=42)
```